SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4 **Total Marks: 50**

Code of Conduct

I K.V Yathish chandra 192524327 (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

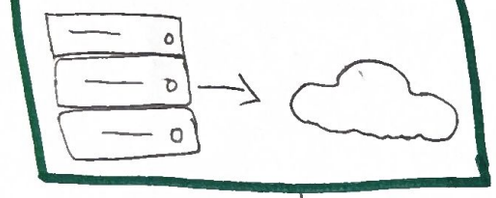
Total Marks:

Cloud Computing Fundamentals

- Definition
  - on-demand of computing resources over the
  - Pay-as-you-go model
- Characteristics
  - On-demand access
  - Resource Pooling
  - Elasticity
  - Scalability
  - Measured service
- Benefits
  - Cost savings
  - High scalability & Flexibility
  - Anytime-anywhere access
  - Automatic updates
  - Disaster recovery
  - Focus on Busines
- Limitations
  - Security & Privacy concerns
  - Internet dependency
  - Vendor lock-in
  - Limited control
  - Data transfer costs
  - Compliance issues

Evolution of cloud computing

Hardware Evolution



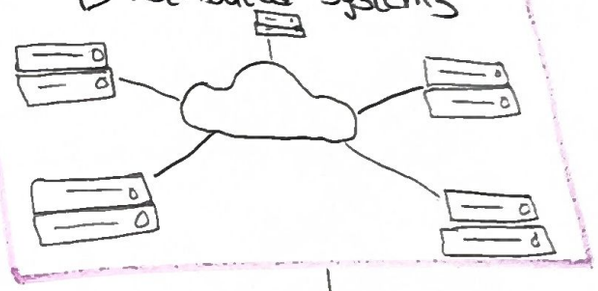
* Mainframes (1st Gen)
* Minicomputers (2nd Gen)
* Personal computers (3rd Gen)
* virtualized / Modern servers (4th Gen)

Internet software Evolution



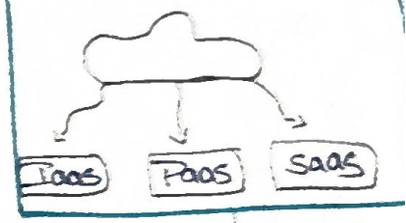
* Static web (1990s)
* Dynamic web (2000s)
* Web 2.0 (Social Media, Collaboration)
* Web 3.0 (cloud, AI, IoT)

Distributed systems



* Multiple computers working together
* Data sharing
* Fault tolerance
* High availability

Service Delivery



* Infrastructure as a se
* Platform as a se
* Software as a se
* Xaas (other s

cloud computing

Service Models

IaaS
Infrastruce as a service)

Provides virtualized Computing resources like servers, storage and networking over the internet

Provider Responbility	user Responbility
* Physical hardware * virtualization * storage & network	* OS * Middleware * Application * Data

Example:
Aws, EC2, Microsoft Azure.

Paas
(Platform as a service)

Provides a complete development and runtime environment for building testing application

Provider Responbility	user Responbility
* Infrastruce * Operating system * Runtime * Middleware	* Application code * Data * configuration * Security

Example
Google APP Engine
Microsoft Azure APP service

Saas
(software as a service)

Provides a complete ready-to-use software applications that are accessed over the internet

Provider Responsibility	user Responbility
* Application * Runtime * Infrastructure	* Data * user setting * Access control

Example
Google workspace
Microsoft 365

Xaas
(Anything as a Service)

A general model where and IT capability is delivered over the cloud as a service

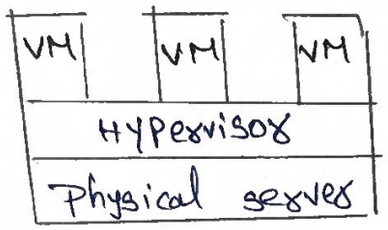
Provider Responbility	user Responbility
* platform * security * Availability	* use the service * Data * Access control

Example
Data base as a service (DBaas)
Backup as a service (Baas)

virtualization and web services as Enables of cloud

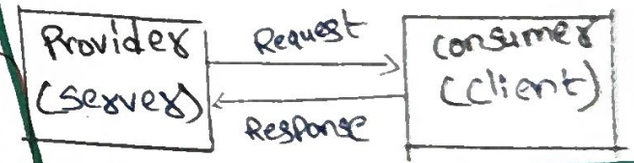
virtualization

* creates virtual machines (VMs) on a physical server
* managed by a hypervisor
* Provides abstraction of hardware resources



web services

* Enables communications between application
* uses standard protocols (SOAP | REST)
* supports Provider and consumer interaction



Cloud Deployment

Cloud Deployment

* Easy integration
* Interoperability
* Faster service delivery

scalability

* Add/remove resource
* Handle varying demand
* Better Performance

12/10/26